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Economics of Household Cooking Using Electricity in Nepal

Sunil Malla
Govinda R. Timilsina
Martin P. Heger

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Abstract

The residential sector is one of the main consumers of energy in Nepal, with cooking being a major end-use. Unprocessed solid biomass fuels are the primary cooking fuels, with approximately 60% of households relying on them for their cooking needs. However, liquefied petroleum gas (LPG), which is entirely imported, is being widely adopted in urban areas. Electricity, which is primarily based on hydropower, a clean domestic energy source, has been used for cooking in less than one percent of households. This paper examines the cost economics of alternative technologies and fuels or their combinations for household cooking across different topographical regions in Nepal from both private and social perspectives. It finds electricity, on average, cheaper than fossil fuels but costlier than biomass fuels from a private perspective. If the costs of local air pollutants, particularly PM2.5, are considered, electricity would be the cheapest option for cooking, except for biogas, which also has minimal external costs. The study also attempts to explore the wider economic benefits of substituting imported LPG with domestic hydropower for household cooking.

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Economics of Household Cooking Using Electricity in Nepal [1]

Sunil Malla, Govinda R. Timilsina and Martin P. Heger [2]

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[2] Govinda R. Timilsina (gtimilsina@worldbank.org) and Martin P. Heger (mheger1@worldbank.org) are, respectively, Senior Research Economist and Senior Environmental Economist, at World Bank Group. Sunil Malla (malla.sunil@gmail.com) is a Short-term Consultant to the World Bank Group.

1. Introduction

The residential sector is the largest energy-consuming sector in Nepal. In 2023, approximately 61% of Nepal's total final energy consumption is attributed to this sector (WECS, 2024a). More than half (54%) of Nepalese households still use unprocessed or traditional biomass (fuelwood, animal dung, and agriculture residue) as a primary source of energy for cooking (NPC, 2024a). In 2019, the World Health Organization (WHO) estimated that 25,447 Nepalese people died prematurely from illnesses[3] attributable to household air pollution (HAP) (WHO, 2024). Women are disproportionately affected more by illnesses associated with respiratory infections, COPD, and lung cancer, attributable to HAP, because they often spend a significant part of their day preparing a meal. It is common knowledge that switching to cleaner fuels or technologies derived from unprocessed biomass for cooking benefits households financially while reducing negative health and environmental impacts.

Recognizing the inefficiency inherent in traditional biomass and cooking stoves, the adverse impacts on human health, and the negative socio-economic consequences (e.g., young girls are forced to collect fuelwoods instead of enrolling them in schools), the government of Nepal has initiated several household-specific clean cooking policies and programs at the national and provincial levels. The primary objectives of the policies and programs were to reduce reliance on traditional biomass for cooking and switch to modern fuels and technologies, such as LPG, electricity, biogas, and improved cookstoves (GoN, 2021; NPC, 2020, 2024b; WHO, 2023).[4] These initiatives have contributed significantly to the adoption of clean cooking

[3] The WHO listed chronic obstructive pulmonary disease (COPD), ischaemic heart disease, stroke, acute lower respiratory infections, and trachea, broncus and lung cancer, as the top five illnesses associated with HAP.
[4] For example, subsidy of up to 50% of the total cost but not exceeding NRs 10,000 for biogas plants with a capacity of 4 m3 or less, and a maximum subsidy of up to 50% but not exceeding NRs 3,000 to 4,000 per improve cookstove (e.g., metallic, *rocket*) per household in rural and peri-urban areas. In the recent budget speech (FY 2023/24), the government has allocated NRs 1.74 billion for the promotion of alternative energy that includes adoption of clean and affordable technologies (biogas, improved and electric cook stoves). In the past budget speech (FY 2022/23), the government has made several decisions, such as i) distribution of one electric stove to one family on grant at the local level under "Women's First Program", ii) distribution of ICS to households that do have access to electricity under "Smoke-free Kitchen Campaign", iii) subsidy for electricity consumption and gradually reducing subsidy for LPG under "Quit LPG Connect Electricity" campaign, iv) substitution of LPG stoves by electric stoves in public offices nationwide, and v) free electric meter and 20 units of electricity per month for poor households. Further, the government of Nepal has set the target of reaching 500 thousand biogas plants, 645 thousand ICS, and 1 million electric stoves for households by 2029 from current (2023) values of 449 thousand, 145 thousand, and 55 thousand, respectively (NPC, 2024b).

in Nepal. For example, households have substantially declined reliance on fuelwood (the primary source of traditional biomass) for cooking, from 64% in 2011 to 51% in 2021 (NSO, 2023). Yet, existing challenges persist, and new ones have emerged. Although the relative size of fuelwood used for cooking decreased, households' reliance on fuelwood continued to rise in absolute terms, from 231 PJ (petajoules) in 1996 to 257 PJ in 2023, representing an 11% increase (WECS, 2006, 2024a).[5] While imported LPG, which increased by a twenty-eight-fold rise between 1996 and 2023 (NOC, 2025), has helped to reduce health impacts from HAP, it has created an economic challenge, particularly a deteriorated trade deficit. Nepal imported NRs 55.6 billion worth of LPG from India last year (FY 2023/24), an equivalent of 3.5% of the country's total import bill, and more than NRs 179.4 billion in the past 3 years (MOF, 2024).

Despite being domestically produced and clean, electricity[6] use for cooking is limited. Only 0.5% of households used electricity for cooking in 2021 (NSO, 2023), far less than the government's goal of achieving 7% in 2021 (NPC, 2020). The situation is such that Nepal has surplus electricity generation capacity during the wet season (June to November), and power producers are compelled to shut down due to a lack of market demand. On the other hand, the country continues to subsidize imported LPG, and biomass for cooking remains prevalent in most peri-urban and rural areas. Considering the increased access to grid electricity,[7] adequate generation at least half of the year (i.e., during the wet season), and switching to electric cooking could benefit the country from economic, health, and environmental perspectives.

However, clean (electric) cooking in Nepal faces complex challenges influenced by socio-economic, behavioral, cultural, and infrastructure barriers. This study analyzes the reasons for the slow adoption of clean cooking in Nepal and investigates the environmental, micro-economic, and macro-economic benefits of switching to electric cooking. A growing body of research is available to analyze the factors contributing to Nepal's slow adoption of clean cooking. Both demand- and supply-side factors, including social, cultural, geographical, political, and economic factors, have contributed to Nepal's heavy reliance on biomass and low

[5] In recent years, however, there is a sign of declining trend in the absolute value of residential fuelwood consumption, e.g., 323 PJ in 2019 to 257 PJ in 2023.

[6] Although some electricity is imported from India, local air pollution due to the imported electricity does not occur in Nepal.

[7] 94% of Nepalese households, 96.2% in the urban areas (excluding Kathmandu valley) and 87.5% in rural areas have access to electricity (NSO, 2024).

uptake of clean energy. The focus areas of these existing Nepal specific studies vary widely from fuelwood use (Amacher et al. (1999); Baland et al. (2018); Kandel at al.2016); Baland et al. (2010); Amacher et al. (1996); Soussan et al. (1991); Fox (1984)] to modern energy use including biogas [Ghimire et al. (2025); Lee and Kwon (2025); Chapagai et al. (2024); Flechtner et al. (2024); Paudel et al. (2023); Bajracharya et al. (2023); Bharadwaj et al. (2022, 2024); Malla (2013, 2021,2022); Acharya and Adhikari (2021); Pokharel and Rijal (2021); Clements et al. (2020); Paudel et al. (2020); Shahi et al. (2020); Bhandari and Pandit (2018); Lam et al (2017); GO et al. (2017)] by households either at the national or local levels, and other wide range of issues, such as socio-economic, behaviors, and cultural factors [KC et al. (2024); (Koirala and Raut, 2024); Robinson et al. (2022); Das et al. (2019); Acharya and Marhold (2019); Joshi and Bohara (2017); Pokharel (2004)] that influence household energy consumption. There are also few studies evaluating various public and government-financed and strategies for improved cookstove and clean and electric cooking programs [Khalifa et al. (2025); Lohani et al. (2025); Ramirez et al. (2024); Shrestha et al. (2020)], and low adoption of electric stoves due to unreliable electricity supply (Koirala and Acharya, 2022). We build on this literature by focusing on supply-side factors, particularly those related to the adoption of electric cooking. While existing recent studies indicate a promising role for electric cooking in Nepal's energy transition [PEEDA (2024); Shrestha et al. (2024); WECS (2022c); PRC (2024)], the economic analysis of household electric cooking across the regions remains insufficiently addressed in the literature. To address these gaps, we calculated the cost of cooking by considering various factors, including raw material input costs, cookstove costs, fuel costs, and labor costs. We also developed a techno-economic model for microeconomic cost analysis. The model calculates the average cost of cooking for the lifetime (i.e., the levelized cost) of various cookstoves and fuels. The analysis also conducts a series of sensitivity analyses on capital costs, O&M costs, and fuel costs, as these are the primary direct cost factors for households in managing their cooking expenses. We also analyzed the social costs, particularly the environmental costs associated with cooking, such as greenhouse gas (GHG) emissions and local air pollution. In addition, the paper also estimates wider economic effects, particularly highlighting savings in fuel import bills, job creation, and value-added in the construction and electricity sectors due to the addition of new hydropower plants to meet cooking fuel demand.

This paper begins with a review of current cooking practices across provinces and selected cities, as well as the pricing structure of cooking fuels. This is followed by the methodologies for analysis, along with a discussion of the data and assumptions. Next, we present the results and conduct a sensitivity analysis. Finally, we conclude with some policy implications.

2. Household Current Cooking Fuel Choices and Prices

Nepalese households use a variety of fuels for cooking, including fuelwood, agricultural residue, animal dung, charcoal, kerosene, LPG, electricity, and biogas.[8] The utilization and patterns of these cooking fuels vary widely across the country depending on geographic location, urbanization, and economic status (**Table 1**). Fuelwood is commonly used for cooking in rural hill and mountain regions, while agricultural residue and animal dung are also used for cooking in the rural Tarai region. In urban Tarai and hill regions, LPG is commonly used for cooking, while electricity (grid) for cooking is limited across the country. Other social, behavioral, and cultural factors also influence households' choice of cooking fuels (Malla and Timilsina, 2014). Some common features of these variations in Nepal include cooking food three or four times a day, using cookstoves for cooking and keeping the room warm in the mountain region where winter temperatures are very low, and cooking outdoors during summer in the Tarai and hill regions. Other fuel-specific common features are that many rural and poor households in the hills and mountains rely heavily on fuelwood for cooking. This reliance is mainly due to its ease of use and free availability from nearby forests, and it requires no processing before use. Agriculture residue is commonly used by rural households in the Tarai (plains) for cooking and animal feed. The Tarai region encompasses most of the country's fertile land, where a variety of crops are cultivated. Animal dung is mainly used by households that own livestock (cattle and buffalo) for cooking in the Tarai and hill. Many rural households rely on inefficient and open cookstoves to cook using these unprocessed biomass fuels.

[8] In country's energy statistics, fuelwood, agricultural residue, and animal dung are categorized as "traditional" biomass fuels, and charcoal, kerosene, LPG, and grid connected electricity as "commercial" fuels. Biogas and off-grid electricity, generated from micro- and pico- hydro and solar, are categorized as "renewables". We use the term "unprocessed biomass" for traditional biomass fuels.

Table 1: Nepalese households' availability/accessibility of cooking/heating energy consumption pattern by urbanization and ecological region

Energy type	Tarai (plains)		Hill		Mountain	
	Rural	Urban	Rural	Urban	Rural	Urban
Fuelwood	●●	●	●●●	●	●●●	●●●
Agri. residue	●●●		●			
Animal dung	●●	●	●			
LPG	●	●●●	●	●●●	●	●
Electricity (grid)		●		●		
Biogas	●	●●	●	●		
Solar (thermal)		●	●	●●	●	●●

Notes: ● is low-level use, ●● is medium-level use, and ●●● is high-level use.
Source: Authors' illustration based on the census data (NSO, 2023, 2024).

LPG, the primary commercial fuel for cooking in Nepal, is widely used by households in both urban and rural areas, including the Tarai and hill regions. In the rural hills and mountains, its use is limited due to a lack of supply infrastructure (e.g., difficulty in transportation) and affordability (i.e., the average household income is low). The other commercial energy (electricity) for cooking is minimal to urban households in the Tarai and hills. It is the same with solar thermal energy for water heating, although it is used by some rural households in the hills and mountains who use their homes as homestays for tourists. Biogas for cooking is location-specific and limited among urban households that own livestock (cattle and buffalo) in the Tarai and hill. Biogas is challenging at high elevations (mountains) due to year-round lower ambient temperatures and lower annual biomass production per area compared to lowlands (Tarai). After the government removed the subsidy in 2014, households no longer use kerosene for cooking. Likewise, very few households use charcoal and bio-briquettes for cooking due to the limited availability of these fuels.

As with many developing countries, households' consumption of clean fuels (LPG and electricity) in Nepal has grown rapidly over the past two decades. For instance, LPG has increased by nearly twenty-three times and electricity by fourteen times in the past 27 years (**Figure 1**). In absolute values, kerosene consumption is significantly reduced at household and national levels, while biogas and solar thermal consumption have increased over the same period (**Table 2**). There is also a sign of households' declining use of agricultural residue and animal dung. However, households' fuelwood consumption in absolute values has not slowed down. Fuelwood remains the dominant cooking fuel in the country's energy mix, representing 79% of households' final energy consumption and 57% of total final energy consumption in 2023 (**Table 2**). The other main cooking fuels (animal dung, agricultural residue, LPG, and biogas) account for only 15% of households' final energy consumption in the same year. Although household electricity has grown rapidly, it is primarily used for lighting and operating non-cooking appliances.

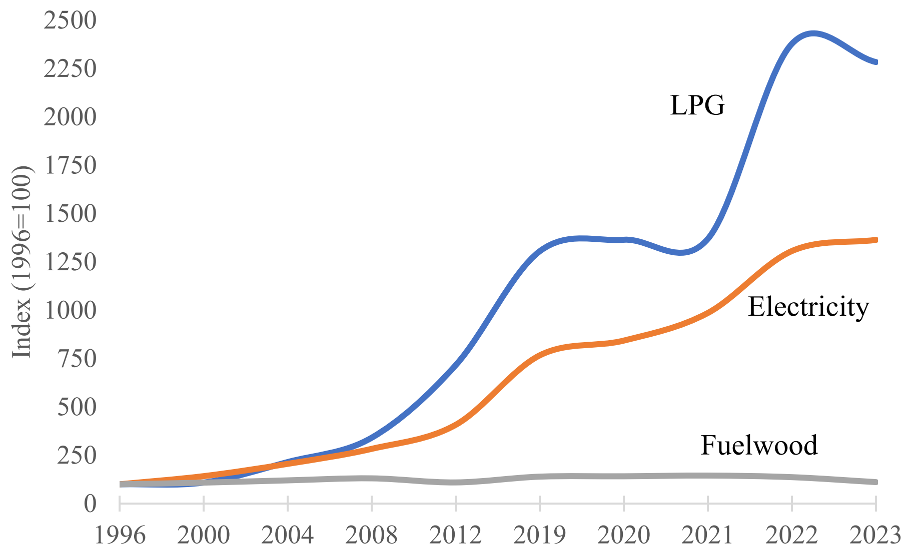


Fig. 1. Residential LPG, electricity, and fuelwood consumption from 1996 to 2023.
Sources: WECS (2010, 2024a)

Table 2: Residential and total final energy consumption (PJ)

Fuel type	Residential		Total	
	1996	2023	1996	2023
Fuelwood	231	257	235	305
Animal dung	18	10	18	10
Agriculture residue	10	9	11	26
Kerosene	6	<1	8	<1
LPG	1	18	1	24
Electricity	1	16	3	39
Biogas	<1	11	<1	11
Others*	<1	3	16	119
Total	268	324	292	532

Note: * Includes aviation turbine fuel, coal, furnace oil, gasoline, diesel, off-grid electricity, solar, and wind.
Sources: WECS (2010, 2024a)

At the province level,[9] households have distinctive cooking fuel choices (**Table 3**). Bagamati, which includes the capital city Kathmandu Valley, has more than two-thirds of households relying on LPG for cooking, the highest among all provinces. In contrast, households in the relatively less developed two western provinces (Sudurpachim and Karnali) depended heavily on fuelwood (70% to 82%) for cooking. Interestingly, 17% of Madhesh households still rely on animal dung, and 9% of Lumbini households rely on biogas for cooking. Except for these two provinces (Madhesh and Lumbini), the proportion of fuels other than fuelwood and LPG (i.e., animal dung, biogas, electricity, kerosene, briquettes, and solar) used for cooking by households is very small, ranging from 9% in Bagmati to less than 1% in Karnali. Although access to electricity in the country has improved in recent years, the use of electric cooking remains low, at less than 1%, in all provinces. The proportion of cooking energy in total residential energy consumption for two provinces (Koshi and Bagmati) is lower (58% and 65%) compared to the other five provinces, which range from 85% for Lumbini to 98% for Kanali. The recent census data shows that provinces also have distinctive household structures. For example, the three centrally located provinces (Bagmati, Gandaki, and Lumbini) together represent 51% of all households, followed by the eastern two provinces (Koshi and Madhesh) at 35%, and by the western two provinces (Karnali and Sudurpachim) at 14%. More than half of all households in each province are in urban areas, with the highest rate (77%) in Bagmati.

Table 3: Province-level residential cooking energy consumption (TJ)

	Koshi	Madhesh	Bagmati	Gandaki	Lumbini	Karnali	Sudurpachim
Fuelwood	13,833	20,011	13,930	13,714	18,622	16,210	22,717
Agriculture residue	27	4,821	1,842	26	170	22	993
Animal dung	551	5,857	-	-	121	-	9
LPG	2,856	2,846	6,729	1,828	3,417	237	504
Electricity	63	74	182	148	110	21	14
Biogas	47	<1	98	306	2,201	3	1,035
Others*	5	13	1	1	34	0	0
Total cooking energy	17,382	33,622	22,782	16,023	24,674	16,493	25,272
% share of cooking energy (%)	58	86	65	83	85	98	93
No. of household (thousand)*	1,192	1,157	1,571	662	1,142	366	577

[9] Administratively, Nepal is divided into seven provinces (Koshi, Madhesh, Bagmati, Gandaki, Lumbini, Karnali and Sudurpachim). Each province is further divided into districts and districts are subdivided into urban- and rural-municipalities. Geographically, Koshi, Bagmati, Gandaki and Sudurpachim provinces stretch across all three ecological regions (Tarai, hill, and mountain), Lumbini streches across Tarai and hill regions, Karnali streches across hill and mountain regions, and Madhesh is only in the Tarai region.

Urban household (%)*	62	73	77	66	55	52	62

Note: * 2021 Census data. Sources: (NSO (2023), WECS (2021, 2022a,b,c, 2023, 2024a).

The government has also recently published clean cooking technology reports for selected cities (Biratnagar, Kathmandu Valley, Butwal, and Pokhara).[10] These reports indicate that households in these cities exhibit distinctive cooking energy use patterns (**Table 4**). For instance, in cities, households mainly used LPG, fuelwood, and electricity for cooking. Over half of the households' cooking energy consumption in these cities comes from LPG, followed by biomass and electricity. Even in cities where electricity is easily accessible, households' use of electricity for cooking remains relatively small, ranging from 6.4% in Biratnagar to 16% in the Kathmandu Valley. LPG is the primary source of cooking fuels in cities, and so is the use of LPG cookstoves. For example, more than 97% of the households use LPG cookstoves in Kathmandu Valley, Butwal, and Biratnagar, and about 87% in Biratnagar. In these cities, households often employ a combination of cooking technologies and fuels to meet their cooking needs. For example, households relying on two or more cookstoves (technology stacking) vary from a high 74% in Pokhara to a low 17% in Kathmandu Valley, while households relying on two or more cooking fuels (fuel stacking) range from 30% in Biratnagar to 76% in Pokhara (**Table 4**).

Table 4: Selected city-level residential cooking energy consumption and fuel/cooking technology stacking

	Biratnagar	Kathmandu	Pokhara	Butwal
Cooking energy consumption (TJ)				
Biomass	68	2,151	423	185
LPG	414	4,096	903	310
Electricity	33	487	246	59
Others	0	2	15	1
Total	515	6,736	1,586	554
Cooking stove stacking (%)				
1 stove	70	83	26	53
2 stoves	26	16	58	43
>2 stoves	4	1	16	4
Cooking fuel stacking (%)				
1 fuel	70	48	24	52
2 fuels	27	47	67	47
>2 fuels	3	5	9	1

Sources: WECS (2022, 2023, 2024a)

[10] Biratnagar is a metropolitan city in Koshi, Kathmandu valley is a metropolitan city in Bagmati, Butwal is a sub metropolitan city in Lumbini, and Pokhara is a metropolitan city located in the Gandaki.

Cooking fuel prices also influence Nepalese households' fuel choices and the frequency of use. Nepal has no indigenous oil resources, and all oil-based cooking fuels (kerosene and LPG) are imported from India. The government-owned Nepal Oil Corporation (NOC) sets the selling (market) price of kerosene depending on international prices (automatic pricing mechanism). In contrast, the selling price of LPG is fixed and subsidized to make it available and affordable to poor and rural households. The government's loss from subsidizing LPG has been estimated at over NRs 11 billion annually in recent years.[11]

The electricity pricing for households is based on the amount of electricity used and the type of meter installed (capacity subscription). The higher the amount of electricity consumed and the meter ratings, the higher the cost of electricity on a per kWh basis. The Nepal Electricity Authority (NEA) is the sole distributor of grid-connected electricity and sets the pricing to consumers in Nepal.[12] In FY 2023/24, the average electricity price for households using between 101 to 250 units (kWh) per month is NRs 9.5 per unit (NEA, 2024). The government has provided nearly free electricity[13] for households consuming 20 kWh or less since 2021 to assist and encourage low-income households to use electricity. For instance, households using less than 20 kWh per month pay only a minimum fixed charge of NRs 30 in 2024, which is reduced from NRs 60 in 2020. After 20 units, households pay progressively higher minimum fixed charges, as well as a charge based on the amount of electricity consumed. Most rural households purchase fuelwood at a very low price or collect it for free from forests to meet their cooking and heating needs. Because of this, it is often difficult for them to justify a shift to cleaner fuels for cooking despite the health and time-saving benefits. Not only this, but some households also foresee the opportunity of selling freely available fuelwood or its by-product (charcoal) in the market for financial gain. Fuelwood prices vary widely across the country. The average market price of fuelwood ranges from approximately NRs 10 to NRs 15 per kilogram, depending on the country's topography (Neupane, 2018; WECS, 2023).

Table 5: Share of households by type of fuel usually used for cooking by region (%)

Region	Municipality	Fuelwood	LPG	Electricity	Others*

[11] https://myrepublica.nagariknetwork.com/news/dual-lpg-price-plan-must-not-harm-the-public-69-66.html
[12] Except for Butwal Power Company supplying a few districts in the western Nepal.
[13] With a minimum charge of NRs 30 per month based on 5 ampere single phase low voltage (NEA, 2024).

Mountain	Rural	89.9	9.1	0.2	0.8
	Urban	77.6	21.3	0.1	1.0
Hill	Rural	85.4	13.8	0.1	0.6
	Urban	33.5	65.3	0.5	0.7
Tarai	Rural	58.3	27.8	0.7	13.3
	Urban	41.5	52.1	0.6	5.8
Kathmandu valley	Urban	4.0	94.8	0.9	0.3
Nepal	Total	51.0	44.3	0.5	4.2
	Rural	75.0	18.9	0.4	5.8
	Urban	39.3	56.7	0.6	3.4

Source: NSO (2023). Note: * Includes biogas, animal dung, and kerosene.

Unlike in many other countries, households in Nepal have limited options for energy sources to meet their cooking needs. Unprocessed biomass, mainly fuelwood, remains the primary cooking fuel for most households. More than half, and as high as 90% of rural households nationwide, rely on fuelwood for cooking (**Table 5**). Even in urban households, cooking with fuelwood remains significant, ranging from about 33.5% in the hill region to 78% in the mountain region. After fuelwood, LPG is the second most popular choice for households as their primary cooking fuel, mainly in urban areas. LPG is prevalent primarily in urban households because it is readily available, convenient, versatile, efficient, and clean burning. Cooking with biogas and electricity is so far insignificant.

Cooking with biomass and fossil fuels also contributes to air pollutant emissions. A recent study (IHME, 2024) estimated that air pollution, primarily fine particulate matter, contributed to 48,500 deaths in Nepal, of which HAP is responsible for approximately 70% (about 34,000 people) of all air pollution-related deaths in 2021 (**Table 6**). Exposure to the HAP is responsible for 49% of all deaths from COPD, 34% of all deaths from stroke, 32% of all deaths from ischemic heart disease, and 31% of all deaths from lower respiratory infections in 2021. After COVID-19, ischemic heart disease, COPD, Stroke, and lower respiratory infections associated with air pollution are the top five causes of death for all ages combined in 2021.

Table 6: Total deaths and percent of deaths from diseases attributable to HAP from solid fuels in 2021

Causes of death	Total deaths (thousands)	Share of HAP (%)	Rank of top 10 causes of death
Lower respiratory infections	6.92	31.0	6
Stroke	17.25	34.0	4
Ischemic heart disease	27.63	32.0	2

Chronic obstructive pulmonary disease	27.54	49.0	3
Trachea, bronchus, and lung cancers	1.33	28.0	-
Others	171.75	1.9	-
Total	252.42 (33.95)*	13.4	-

Source: IHME (2024). Note: * The figure in parentheses shows the total deaths due to HAP.

3. Methodology and Data

3.1 Methodology and data to estimate the private costs

We employ a levelized cost of cooking (LCOC) approach to evaluate a household's average cost of cooking across various fuel sources and cooking technologies. The results would help households understand the overall cost efficiency of their cooking practices using different fuel-cookstove combinations. The LCOC framework used in this study is illustrated in **Figure 2**. It involves three steps of estimation for different topographical regions (Tarai, hill, and mountain), as well as the Kathmandu Valley, and for both urban and rural areas in Nepal. The first step is to estimate the useful energy requirements (or energy service) for households to cook. The useful energy refers to the energy available to households to satisfy their cooking needs. It is independent of the cooking devices or fuels used. Still, it depends on the household's cooking events or habits (e.g., the number of meals prepared per day), the number of household members, and geographical location (altitude). Depending upon the economic conditions of the households, heating events[14] may vary. For example, due to suppressed demand, the number of meals prepared by low-income households is likely to be lower than that of high-income households.

In contrast, households in cold-temperature (high-altitude) areas are more likely to consume a greater quantity of fuel than households in warm-temperature (low-altitude) areas for cooking the same type of food. Accounting for these differences is crucial in analyzing cooking fuel consumption in a country like Nepal, but it is often overlooked or poorly represented in the literature. We first estimated the available aggregated or location-specific information from the

[14] Heating events cover either a meal (breakfast, lunch, dinner) or other heating purposes such as heating water or preparing snack (Scott, 2021). A single heating event may include several dishes and/or heating water as part of that meal.

literature to capture these differences. We then differentiated these values by heating events, using both urban and rural areas, as well as topography (mountain, hill, and plains) (see **Table A.2**, Appendix A).

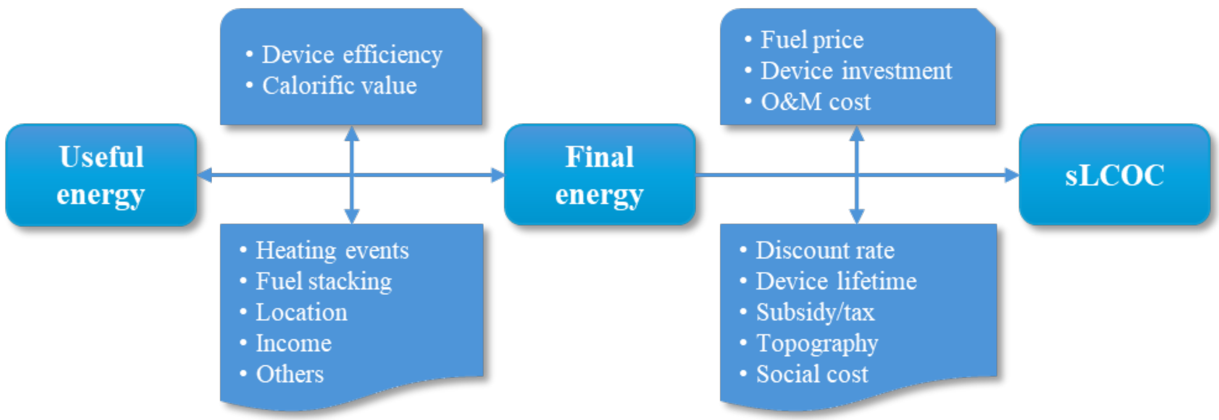


Fig. 2. Framework for estimating household level levelized cost of cooking (LCOC)
Note: Social costs (i.e., health impact costs and opportunity benefits (time saved)) are excluded from financial costs.

Several data types are relevant to this study, such as cooking device-specific, average, and generic data. For this study, data is collected from secondary sources. One of the main secondary sources of data and relevant information is the recently conducted National Population and Housing Census (NPHC) 2021. Other secondary data are collected and compiled through comprehensive desk study, review of relevant peer-reviewed articles, and government and non-government published documents, reports, and articles. **Tables 7-10** summarize the key assumptions and technical and pricing data (including capital and operating and maintenance (O&M) costs of cooking devices, their thermal efficiencies, economic lives, and fuel prices) used in the calculation of levelized costs. The input data used to estimate the wider economic impacts of substituting LPG with electricity for household cooking in Nepal are provided in the results section later, making it easier for readers to understand the estimation of these effects.

Table 7: List of key parameters and data considered in the study

Study period	2021
Area coverage	Mountain, Hill, and Tarai are differentiated by rural and urban areas; and Kathmandu
Cooking fuel/technology	Fuelwood, biogas, Kerosene, LPG, Electricity

Household size	4.38 (mountain-rural); 4.23 (mountain-urban); 4.14 (hill-rural); 3.90 (hill-urban); 4.99 (Tarai-rural); 4.63 (Tarai-urban); 3.89 (Kathmandu valley); and 4.37 (Nepal)
Conversion factor	1 kWh = 3.6 MJ; 1 toe = 41868 MJ; 1 kcal = 0.004184 MJ
Data for economic spillover effects	Input-output tables or Social Accounting Matrix; sectoral employment statistics; Other economic data

Sources: NSO (2023); WECS (2024a).

Table 8: Thermal (heat) efficiency, initial investment cost, and lifetime of different cookstoves in Nepal

Cookstove	Thermal efficiency (fraction)	Initial investment (thousand NRs)	Lifetime (year)
Traditional cookstove (TCS)	0.12	0.00	1
Improved cookstove (ICS)	0.24	0.80	4
Metallic cookstove (MCS)	0.18	1.13	5
Biogas	0.48	85.25	15
Kerosene	0.45	3.50	10
LPG	0.55	7.25	10
Electric			
Hot plate	0.70	4.50	10
Infrared	0.70	6.25	10
Induction	0.78	9.00	10
Electric pressure cooker (EPC)	0.86	13.00	10

Sources: (AEPC, 2016; Gautam et al., 2021; Malla, 2022; WECS, 2022, 2023)

Table 9: Cooking fuel market price in Kathmandu Valley and average net calorific value (NCV)

Fuel type	Unit	Price (NRs/unit)	Price (NRs/MJ)	Average NCV (MJ/kg)
Fuelwood	kg	15.00	0.90	16.75
Biogas	m^3	0.00	0.00	17.39
Kerosene	liter	152	2.85	35.00
LPG	cylinder	1800	2.75	46.10
Electricity	kWh	9.72	2.70	3.60

Note: The conversion factors used are as follows: 1 liter of kerosene is equivalent to 0.81 kg, one cylinder of LPG is 14.2 kg, and the exchange rate is 1 US dollar = NRs 125.
Sources: (NEA, 2023a,b; NOC, 2025; Kathmandu Post, 2015; WECS, 2010, 2023).

Table 10: Fuel price index by region used in the study (Index, Kathmandu Valley price = 100)

Fuel type	Nepal		Mountain		Hill		Tarai	
	Rural	Urban	Rural	Urban	Rural	Urban	Rural	Urban
Fuelwood	0	50	0	60	0	40	0	50
Kerosene	128	117	150	150	125	100	110	100
LPG	128	117	150	150	125	100	110	100
Electricity	100	100	100	100	100	100	100	100

Source: Authors' assumption in consultation with experts.

The second step is to estimate the final energy demand. This step requires information, such as the cooking device's thermal efficiency and the share of device-specific fuel use, to assess the household's energy demand for cooking in a common unit, e.g., in giga joule (GJ) for each technology-fuel combination. For example, if the traditional stove efficiency is 10% and the improved stove efficiency is 25%, the fuel savings should reflect a factor of 2.5. In this step, fuel stacking is implicitly considered. This is important because many Nepalese households use multiple fuels for cooking.[15] Using the net calorific value of each cooking fuel, the final energy demand for each fuel is then converted into natural units, e.g., kg for fuelwood, LPG and biogas, and kWh for electricity. It is essential to note that due to transformation losses at the point of use, the amount of final energy is greater than the corresponding useful energy demand for all cooking devices. Further, within different cooking devices, electric cooking devices often have higher conversion efficiency than devices using other fuels, meaning that for the same amount of end-use energy services (e.g., heat for cooking), less final energy (e.g., MJ or kWh) is required for electric cookstoves as compared to those using other fuels (e.g., biomass, kerosene, LPG).

The third and final step is to estimate the LCOC. It includes the costs of cookstoves (e.g., initial investment, maintenance, and repair costs) and the costs of energy sources used for cooking but excludes the costs of food items or other cooking ingredients.

The levelized costs of cooking from the private perspective ($LCOC_i$) for each cooking technology (cookstoves) type (*i*) in Nepal includes operational (*OpEx*) and capital (*CapEx*) expenditures

$$LCOC_i = OpEx_i + CapEx_i \tag{1}$$

The annual operational expenditures on cookstove type i ($OpEx_i$) represent the ongoing cooking costs required to prepare a standard food. It includes fuel or energy costs (*CE*) and operation and maintenance costs (*CO&M*)

$$OpEx_i = CE_i + CO\&M_i \tag{2}$$

The different sources of energy used for cooking considered in this study include combustible fuels (e.g., fuelwood, animal dung, agriculture waste, charcoal, kerosene, and LPG), biogas, electricity, and solar energy. The energy sources used are specific to the type of cookstove, for example, LPG for LPG cookstoves, electricity for hot plates, infrared and

[15] For example, about 8% of the Nepalese households use the combination of fuelwood and LPG, and 4% of households use LPG, fuelwood, and dung, for their cooking needs (World Bank, 2019).

induction cookstoves, electric pressure cookers and rice cookers, and unprocessed biomass for traditional or improved cookstoves.

The capital expenditures ($CapEx_i$) include the annualized value of initial capital investment, such as the cost of equipment and installation costs. The initial capital expenditure is annualized so that it can be added with annual energy and O&M costs to get the total yearly costs. The annualized capital expenditure on cooking is estimated as follows:

$$CapEx_i = C_i \times \frac{r(1+r)^{ni}}{(1+r)^{ni}-1} \tag{3}$$

Where C_i is the initial capital costs, including equipment and installation costs of cookstove type *i*; *r* is the interest rate, and n_i is the economic life of cookstove *i*.

The private cost of cooking depends on the quantity of fuels consumed by households for cooking, which varies widely by location, cuisine, culture, and many other factors. Several laboratory- and field-based methods are used to estimate fuel consumption or savings and evaluate cookstove performance and quality for cooking.[16] Among these methods, thermal energy output (TEO), control cooking test (CCT), kitchen performance test (KPT), water boiling test (WBT), and cooking diaries (CD) are commonly used.[17] The selection of methods for calculating the private cost of cooking depends on the trade-off between the preciseness of results versus the data collection costs (Lee et al., 2013). In this study, we utilize information collected from the cooking diaries method (Poudel et al., 2023) and surveys of residential and commercial energy consumption nationwide (WECS, 2022b, 2023, 2024a). We then use the following general equation to estimate the LCOC for each cooking device *i* using fuel type *e*.

$$LCOC_i = CapEx_i + CO\&M_i + CE_i \tag{4}$$

[16] https://cleancooking.org/research-evidence-learning/standards-testing/protocols/

[17] The TEO rates the thermal capacity of the cookstove along with other parameters to estimate the fuel savings (GS, 2016). The CCT is a field test that measures stove performance in comparison to traditional cooking methods when a cook prepares a pre-determined local meal (Bailis, R., 2004). The KPT is the principal field–based procedure that measures household fuel consumption under typical household and stove usage conditions (Bailis et al., 2018). The WBT is a laboratory-based test that measures how efficiently a stove uses fuel to heat water in a cooking pot and the quantity of emissions produced while cooking (GACC, 2014). In the cooking Diaries method, a small number of cooks are typically asked to cook as normal for a couple of weeks (baseline phase) and then to swap using electric cooking appliances to cook as much of their menu as possible during a transition phase. During the study, cooks record all the dishes cooked, and measure the energy used (Leary et al., 2019).

$$E_i(t) = EP_{,e} \times \left(\frac{1}{CV_e} \times \frac{UE(t)}{\eta_i}\right) \tag{5}$$

The final energy cooking cost (**Equation 5**) is estimated by multiplying the price of the final energy used for cookstove *i* and the quantity of final energy used [$\left(\frac{1}{CV_e} \times \frac{UE}{\eta_i}\right)$]. The quantity of final energy in a natural unit (e.g., kg or liter or kWh) is estimated by dividing the useful energy requirement for cooking (*UE*) with the calorific value of fuel type *e* (CV_e) and the thermal efficiency of the cooking device *i* (η_i). The useful energy required for cooking is the same across different cooking devices, but it varies across the country's regions when cooking and heating events are taken into account.[18] The data and data sources for estimating final cooking energy use are summarized in Tables 7-10 above and Table A.2 in the Appendix.

Households often use multiple fuels for cooking. To capture this, we also estimated the levelized cost of cooking with fuel stacking ($LCOC_{FS}$) using the following equation.

$$LCOC_{FS} = \sum esh \times LCOC_i \tag{6}$$

Where *esh* is the share of fuel type *e* (fuelwood, dung, kerosene, LPG, electricity, biogas, and others) used for cooking. Note that $\sum esh = 1$. The $LCOC_i$ is the levelized cost of cooking associated with device *i* that uses fuel type *e*. We assume the cooking devices used for different fuel types are as follows: ICS for fuelwood, TCS for dung and other sources, infrared for electricity, kerosene stove for kerosene, and LPG stove for LPG and biogas.

3.2 Methodology to estimate the external costs

We then estimated the external cost of cooking by adding health impact costs (due to household air pollution) and the social costs associated with CO_2 emissions. These external costs associated with cooking include environmental costs, such as HAP, greenhouse gas emissions, black carbon, health costs, and other societal impacts. We use the following equation to estimate

[18] In general, the number of cooking/heating events vary across the country depending on the topography and households' income. The estimated weighted average useful energy for different regions are provided in Appendix A (Table A.2).

the social costs of CO_2 emissions and the health damage costs associated with PM2.5 from cooking.

$$C_{i,p}^{S} = EF_{i,p} \cdot FE_i \cdot SC_p \tag{7}$$

In this study, the estimated externality costs are limited to environmental and health damage costs ($C_{i,p}^{S}$) which is the product of the social cost of pollutant p from cooking using device i (SC_p) and the corresponding pollutant p (i.e., CO_2 and PM2.5) emissions (**Equation 7**). The CO_2 emissions are estimated by multiplying the CO_2 emission factor ($EF_{i,CO2}$) and final energy consumption for cooking using device i (FE_i). Likewise, we estimated health damage costs associated with PM2.5 ($C_{i,p}^{S}$) using a similar approach as that for CO_2 emissions. The social cost values for CO_2 and PM2.5 are collected from the literature. We calculated these social costs of cooking in different topographical regions for alternative cooking technologies. The social cost is the total cost incurred by society, encompassing both private and external costs. We estimated the PM2.5 and CO2 emissions using the emission factors (**Table 11**) and the final household's final energy consumption provided in Table A.3. We then used the social cost factor (**Table 11**) for each of these pollutants by multiplying the total final energy consumption to calculate the external costs.

Table 11: $PM_{2.5}$ and CO_2 emission factors from cookstoves and social cost factors

Pollutant type	Emission factor (g/MJ of final energy use)						Social cost (US$/kg of pollutant)
	TCS	ICS	MICS	Biogas	Kerosene	LPG	
$PM_{2.5}$	0.40	0.35	0.35	0.01	0.05	0.01	86.85
CO_2	-	-	-	-	67	64	0.04

Sources: Weyant et al. (2019); Yawale et al. (2023); EPIC INDIA (2018).
Notes: ICS TCS is a traditional cookstove, ICS is an improved cookstove, and MICS is a mettalic improved cookstove.

3.3 Methodology for estimating wider economic benefits

Here, we present the methodology for assessing the economic spillover effects of substituting imported LPG with domestically produced hydropower for residential cooking. It would be more logical to assess macroeconomic impacts dynamically, forecasting future energy demand for household cooking in the baseline and replacing LPG with electricity over time. A complex model, such as a computable general equilibrium (CGE) model, would be more appropriate for this purpose. However, forecasting future energy demand for household cooking is beyond the scope of this study, and the lack of data constrains the use of the CGE model. Therefore, we used a simple model based on input-output coefficients and macroeconomic ratios (e.g., fixed shares of sectoral value added in GDP and sectoral labor productivities). The impact assessments presented here are for a hypothetical case in which we investigate the impacts of replacing all LPG consumed in the year 2023 with electricity. This is purely a hypothetical case, as replacing a large amount of LPG cannot be done in a short period. It is simply an indicative analysis for a what-if case. For this purpose, we first calculate the amount of electricity needed to provide the same level of cooking energy that LPG consumed in the year 2023:

$$ELEC = LPG \cdot CF \tag{8}$$

Where *ELEC* is the electricity consumption in GWh equivalent to LPG consumption expressed in petajoule (PJ), and *CF* is the factor to convert PJ to GWh, which is standard with a value of 277.778 (IEA, 2004). Electricity generation (*ELEG*) to meet the demand (expressed in GWh) would be higher than consumption due to transmission and distribution loss (*tdl*). It is calculated by adjusting the loss as follows:

$$ELEG = \frac{ELEC}{(1-tdl)} \tag{9}$$

The installed capacity in MW (*ELCAP*) to be added into the power system to meet the new demand caused by the LPG substitution is calculated as follows:

$$ELCAP = \frac{ELEG}{(\frac{PLF}{100}*24*\frac{365}{1000})} \tag{10}$$

Where *PLF* is the average plant capacity factor of hydropower plants in Nepal in 2023. Annual investment (*AINV*) needed to build the additional hydropower plants to meet the increased electricity demand due to the LPG substitution with electricity for household cooking is calculated as follows:

$$AINV = \frac{ELCAP.CAPC*1000}{CP} \tag{11}$$

Where *CAPC* is the average capital cost per kW to build a hydropower plant in Nepal, and *cp* is the average construction time of a hydropower project. Annual labor payments during the construction of a hydropower project (*ALPC*) are calculated by multiplying total construction investment by wage & salary to gross output ratio (*LPGORC*) for this sector available from an I-O table or social accounting matrix (SAM):

$$ALPC = AINV * LPGORC \tag{12}$$

The number of jobs or employment (person-year) created in the construction of hydropower (*NLC*) is calculated as follows:

$$NLC = \frac{ALPC * ER}{WRC} \tag{13}$$

Where *WRC* is the average wage rate in the construction sector (NRs per year), and *ER* is the exchange rate. The annual value added created by the substitution of LPG with electricity through the construction sector during the construction period (*VAC*) is:

$$VAC = AINV * VAGORC \tag{14}$$

Where *VAGORC* is the value added to the gross output ratio for the construction sector obtained from the SAM.

Besides the construction of the hydropower power plants, jobs and value-added are also created through the operation of the power plants. The annual number of employees for the operation of the power plants (*NLE*) is calculated as follows.

$$NLE = \frac{ELEC * LPELR}{WREL} \tag{15}$$

Where *LPELR* is the ratio of wage payments to electricity sales (NRs/GWh) of Nepal Electricity Authority in 2023, and *WREL* is the average annual wage rate for the electricity sector in 2023.

The annual value added created by the operation of electricity plants built to substitute LPG with electricity during the operation period (*VAEL*) is:

$$VAEL = ELEC * VAGOREL * VAEL \tag{16}$$

Where *VAGORC* is the value added to the gross output ratio for the electricity sector, obtained from SAM, and *VAEL* is the electricity sector value added per GWh of electricity sold calculated from electricity statistics and the SAM.

4. Results from Economic Analysis of Household Cooking

4.1 Private cost of household cooking

The cost of cooking meals in Nepal can vary depending on several factors, including the ecological region, the type of ingredients used, energy costs, cookstove costs, and individual preferences. In the following section, we focus on the household's levelized cooking costs, which is the average annual cost of cooking using different fuels and technology combinations.

The results show that, excluding traditional cookstoves (TCS), on average of all regions, fuelwood (ICS and MICS) is the most economical cooking technology for households, followed by biogas, electricity (hot plate, induction, and EPC), LPG, infrared, and kerosene (**Table 12**). However, the levelized cost of cooking with different energy types varies widely across the regions. For example, fuelwood is collected for free in many rural areas across the country, making it the least costly option for cooking, as seen in the cases of TCS, ICS, and MICS (**Table 12**). In contrast, cooking with fuelwood in most urban households (in the mountains and Tarai) is more expensive than cooking with biogas and electricity. Despite this, approximately 47% of urban households, excluding those in the Kathmandu Valley, still rely on fuelwood as their primary source of cooking fuel (NSO, 2024). Other examples include cooking with fuelwood, which is cheaper for households in urban hills than in urban mountain and Tarai regions. This variation is primarily due to the different household sizes in these regions and the corresponding relative fuel prices. The household size is smaller in urban areas on the hill (3.9) than in mountain (4.31) and Tarai (4.63) urban areas. Also, fuel prices are relatively higher in the mountains than in the hill regions. The cost of cooking using biogas fuels is the same across regions because the cost of biogas fuel is zero.

Table 12: Levelized cost of cooking by device type (thousand NRs/household/year)

	Mountain		Hill		Tarai		Kathmandu Valley	Nepal		
	Rural	Urban	Rural	Urban	Rural	Urban		Average	Rural	Urban
TCS	0.0	18.7	0.0	11.5	0.0	17.1	28.7	9.8	0.0	15.9

ICS	0.3	9.7	0.3	6.1	0.3	8.8	14.6	5.2	0.3	8.3
MICS	0.4	12.9	0.4	8.1	0.4	11.8	19.5	6.9	0.4	11.0
Biogas*	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1
Kerosene	37.9	40.6	26.8	25.3	28.4	29.9	25.2	31.3	29.9	32.3
LPG	31.0	33.2	22.3	21.1	23.5	24.7	21.0	25.8	24.7	26.7
Hot plate	16.2	17.3	13.7	16.0	16.4	18.8	15.9	16.5	14.9	17.6
Infrared	16.6	17.7	14.2	16.5	16.8	19.2	16.4	16.9	15.3	18.0
Induction	15.7	16.7	13.6	15.6	15.9	18.1	15.6	16.0	14.6	17.0
EPC	15.5	16.4	13.5	15.4	15.7	17.6	15.3	15.8	14.5	16.6

Note: * The cost of cooking with biogas is same across all regions. This is manily due to assuming the cost of biogas fuel is zero, and not differentiating the costs of of building biogas plant and the availability or accessibilty to raw maeterials (animal dung) in producing biogas across the regions. Biogas for cooking is location-specific and it may not be feasible in all regions.

The annual cost of cooking is approximately NRs 24.7 thousand using LPG and NRs 18.1 thousand using induction stoves (electricity) in urban Tarai (**Table 12**). For comparison, WECS (2024b) reported that the average annual cost of cooking is NRs 26.9 thousand using LPG and NRs 13.3 using an electric stove for Biratnagar city, located in the urban Tarai region. Likewise, WECS (2022c) reported that the annual cost of cooking using LPG is approximately NRs 19.3 thousand for Butwal City (urban Tarai region) and NRs 19.9 thousand for Pokhara City (urban hill region). The Nepal Living Standards Survey IV (NSO, 2024) reported that the household's annual fuel expenditure, which also includes non-cooking activities (including fuels used for personal transportation, lighting, and heating), is approximately 61.4 thousand for urban regions and 52.3 thousand for rural regions. In our study, the results show that the average annual cost of cooking in rural and urban regions in Nepal is approximately 14.5 thousand and 16.6 thousand, respectively (**Table 12**). When compared to the NLSS IV survey, these figures translate to approximately 28% of all fuel expenditure being used for cooking in rural regions and 27% in urban regions.

Several factors, including geographic location, economic conditions, and cultural preferences, contribute to households' reliance on fuelwood for cooking. Access to modern energy sources, such as LPG or electricity, can be challenging in remote and rural areas due to rugged terrain, inadequate infrastructure, and high distribution and transportation costs. Furthermore, purchasing and maintaining modern cooking appliances and fuels, such as LPG, can be expensive in rural areas. As a result, households often resort to using readily available and cost-effective fuelwood. Additionally, traditional cooking practices in Nepal typically involve using open fires and simple stoves that are compatible with fuelwood.

However, even if fuelwood is the cheapest cooking option, adding health-related costs associated with HAP from its use and the opportunity cost of time spent collecting it would likely make it the most expensive cooking solution. Additionally, costs related to the lack of availability, affordability, and reliability of cooking with biogas, LPG, and electricity are not taken into account. Furthermore, biogas may not be feasible in some regions of the country, while the supply of imported LPG is also constrained due to a lack of infrastructure, particularly transportation, in rural and remote areas. Kerosene is found to be the least economical cooking solution. Following the removal of subsidies in 2014, the use of kerosene has declined dramatically in the country. The recent National Housing and Population Census 2021 finds that less than 0.05% of households use kerosene for cooking. Although biogas is a clean and relatively economical option, its use has many limitations. For example, its initial investment is huge, and it requires 36−45 kg dung, the equivalent of 4 cows, to produce 1.6 m^3 of biogas per day, which is suitable for cooking meals for a household with five people (AEPC, 2015). Considering all these issues, our LCOC analysis indicates that cooking with electricity is likely to be a promising clean cooking option.

As cooking in Nepal changes along the energy ladder with topography regions and increased income, households rarely switch completely from one fuel to another. Same households often use fuel stacking for cooking – a combination of multiple stoves and fuels. With households using fuel stacking for cooking, as presented in **Table 5**, we find the total cost of cooking in urban areas is higher than in rural areas in all regions (**Figure 3**). In particular, the cost of cooking is the highest for the urban households in the capital city (Kathmandu Valley), followed by urban households in the Tarai, hill and mountain regions. This result is expected for two main reasons. First, most urban households use LPG, which is relatively expensive compared to fuelwood and electricity in terms of cooking energy costs per household. Second, the household's share of LPG among different fuel uses is highest in the Kathmandu Valley and lowest in the mountain region. The cost of cooking in rural areas is generally lower than in urban areas across all regions, primarily due to the high share of fuelwood consumption in rural areas, which is freely available.

Fig. 3. Levelized cost of cooking (LCOC) with fuel stacking (thousand NRs/household/year)[19]

4.2 External costs of household cooking

Unprocessed solid biomass fuels, such as fuelwood, require a large mass for cooking relative to the energy delivered due to their lower heating value. Because of this, these fuels are responsible for high levels of household air pollutant emissions. Regardless of the type of stove used for cooking, combustion of fuels, excluding electricity, produces certain pollutants, such as fine particulate matter ($PM_{2.5}$), carbon monoxide, black carbon, and CO_2.[20]

The local air pollutants (CO and $PM_{2.5}$) are the major products of incomplete fuel combustion, and they are closely related to indoor air quality and human health. We estimated

[19] The estimated household's average annual cost of cooking by cooking devices are NRs 9,835 for TCS, NRs 5,228 for ICS, NRs 13,101 fo biogas, NRs 31,307 for kerosene, NRs 25,838 for LPG, and NRs 16,933 for infrared electric stoves. We used these values and the the share of households by type of fuel usually used for cooking by region (Table 5) to estimate the levelized cost of cooking with fuel stacking. We use the comination of device/fuel as follows: TCS/others ICS/fuelwood, electricity/infrared.

[20] Some studies (Tormey and Huntley, 2023); Seals and Krasner, 2020; O'Leary et al., 2019) find that even when cooking on electric stoves, the $PM_{2.5}$ is produced by the vaporizing of grease particles from the cooking pan.

selected household air pollutants from cooking fuel combustion by regions across the country (**Table 13**). We find significant variation in the magnitude of air pollutants across different regions. For example, the $PM_{2.5}$ emissions vary from a low of 0.3 kg per household in the valley to as high as 5.1 kg per household in the rural mountain region. This is because rural mountain areas primarily use fuelwood, which emits higher levels of $PM_{2.5}$. Since fuelwood use is common and dominant in rural areas, CO emissions are relatively higher in rural areas than in urban areas (**Table 13**). The emissions of black carbon (BC) also show similar trends. BC emission is closely related to the carbon content in the fuel; i.e., higher carbon fuels produce more BC during the combustion process. In contrast, CO_2 emissions are higher in urban areas than in rural areas because these emissions are associated with fossil fuels (LPG and kerosene) commonly used in urban areas.

Table 13. Annual emissions of selected household air pollutants from cooking by region[21] in 2021 (kg/household/year)

	Mountain		Hill		Tarai		Kathmandu Valley	Nepal		
	Rural	Urban	Rural	Urban	Rural	Urban		Average	Rural	Urban
$PM_{2.5}$	5.1	4.7	4.0	1.9	4.8	3.4	0.3	3.4	4.4	2.8
CO	62	58	49	25	60	43	5	42	54	35
BC	0.7	0.6	0.6	0.3	0.6	0.4	<0.1	0.5	0.6	0.4
CO_2	42	105	53	294	128	277	424	206	79	281

Air pollution is recognized as an important risk factor for several health-related diseases, including COPD, ischemic heart disease, stroke, and lung cancer, and in the past ten years, mortality rates associated with air pollution have been increasing substantially (WHO, 2023). The WHO estimated that the number of deaths due to diseases related to exposure to household (indoor) air pollution is 25,069, for a rate of 88 per 100,000 population, which is much higher than deaths due to exposure to ambient (outdoor) air pollution (14,582 with a rate of 51 per 100,000 population) in Nepal in 2019 (WHO, 2024). COPD, ischemic heart disease, and stroke are the top three leading causes of death for females in 2019.[22] These health effects result in economic and welfare losses, including death, lower life expectancy, illness, increased healthcare spending, and reduced productivity. The estimated economic loss due to premature death

[21] These annual emssions are estimated by multiplying the cooking device specific emission factor for each pollutant type and the final energy consumption of households that varies across the region. The household's final energy conusmption is esitmated as the product of useful energy required and the share of fuel/cookstove for households in each region, and then multiplying by corresponding cookstove efficiency.

[22] World Health Organization 2024 data.who.int, Nepal [Country overview]. (Accessed on 21 March 2024).

attributable to HAP was NRs 39.8 billion in 2019, equivalent to 0.93% of the country's total GDP in that year.[23]

We also estimated the external costs associated with $PM_{2.5}$ and CO_2 emissions from cooking in the country (**Table 14**). We find that the external costs of $PM_{2.5}$ under fuel staking are much higher for rural households than for urban households. This is mainly due to rural households using more unprocessed biomass fuels than urban households. However, when cooking with a single fuel source (e.g., fuelwood or LPG), the external costs of PM2.5 are relatively lower for rural households compared to urban households. This is primarily due to urban households cooking more frequently than rural households, which requires more fuel for cooking. Likewise, we find that the external costs associated with CO_2 emissions are higher for urban households than for rural households under the fuel stacking and cooking with one fuel case, mainly due to urban households using more fuel for cooking than rural households.

Table 14: External costs of PM2.5 and CO_2 emissions of cooking in Nepal (thousand NRs per household per year)

		Mountain		Hill		Tarai		Kathmandu valley	Nepal
		Rural	Urban	Rural	Urban	Rural	Urban		
PM2.5	Fuel stacking	58	53	46	22	54	38	3	38
	Fuelwood (ICS)	63	68	53	63	64	75	63	65
	LPG	0.8	0.8	0.6	0.8	0.8	0.9	0.8	0.8
CO2	Fuel stacking	0.2	0.5	0.3	1.5	0.7	1.4	2.2	1.1
	LPG	2.4	2.5	2.0	2.3	2.4	2.8	2.3	2.4

Note: We use the social costs of $PM_{2.5}$ as US$ 86.85/kg and CO_2 as US$ 0.04/kg as reported in EPIC INDIA (2018). The exchange rate used is 1 US dollar = 130 Nepalese rupees.

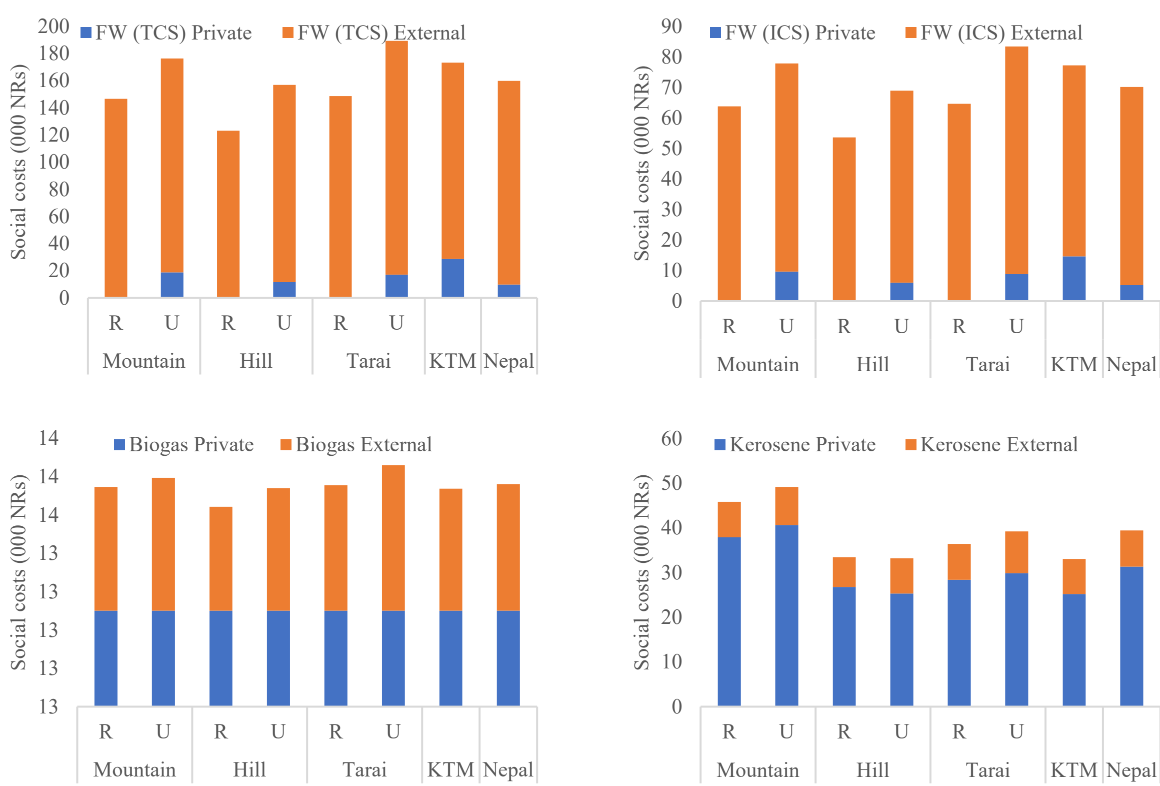
Cooking with electricity has several health benefits compared to cooking with fuel combustion, such as unprocessed biomass and LPG. Some health advantages associated with electric cooking are improved indoor air quality, reduced respiratory health risks, such as COPD, asthma, and respiratory infections, prevention of HAP, mitigation of indoor smoke exposure, and prevention of burns and injuries, especially for households with children or individuals who may

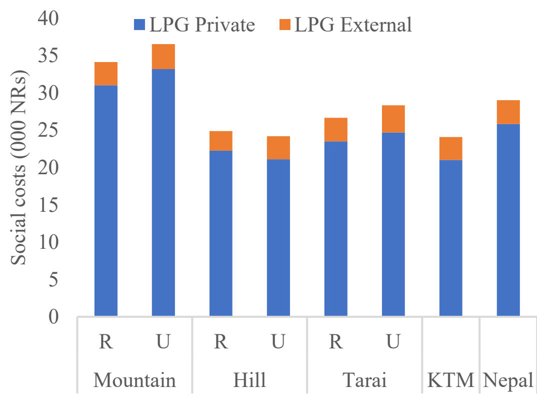
[23] Based on the estimated value of India. In India, the average number of deaths attributable to HAP is 606,890 and the economic loss due to deaths and morbidity attributable to HAP is US$ 13,300 million (Various, 2021). This translates into an average value of economic loss of US$ 21,915 per death in India. Assuming similar figure, the corresponding value of economic loss for Nepal is US$ 12,692 per death, adjusted with per capita GDP in 2019 (GDP per capita is US$ 1185.68 for Nepal, and US$ 2047.23 for India in 2019). We use exchange rate of 1 US$ is NRs 125.

be more vulnerable to accidents. The other benefits of cooking with electricity include safety and convenience, flexibility, and versatility.

4.3 Social costs of household cooking

Figure 4 presents the total costs to society (including private and external costs) of cooking in Nepal for different topographical regions. Although fuelwood incurs the highest external costs, it remains the most affordable option for cooking in rural areas because it is readily available for free from nearby forests, and alternative cooking options are limited. Many urban households commonly use LPG due to its availability, convenience, efficiency, and cleanliness compared to fuelwood. Despite the private cost of cooking with LPG being significantly higher than that of cooking with electricity in rural and urban areas (**Figure 4**), it is widely used due to its easy access and lack of alternative cooking options.





Note: FW is fuelwood, TCS is traditional cookstove, ICS is improved cookstove, R is rural, U is urban, and KTM is Kathmandu Valley.

Fig. 4. Social costs (private and external) of cooking with selected fuel/technology by region

Between cleaner cooking fuels, electricity is more cost-effective than LPG. For example, compared with LPG cooking, private electricity (infrared) cooking costs are 22% cheaper in Kathmandu and 35% cheaper in Nepal (**Figure 4**). While electric cooking is a promising opportunity due to its environmental and health benefits, convenience, and efficiency over LPG and fuelwood cooking, it faces some key challenges (e.g., lack of electricity supply infrastructure, high costs of electric cook stoves and appliances, and other cultural and behavioral factors) that are limiting its widespread use in the country.

4.4 Wider economic effects of substituting LPG with electricity

As Nepal's electricity sector rapidly expands, the potential for replacing imported LPG with electric cooking solutions is gaining attention. This shift is driven by economic, environmental, and health considerations. Households' electric cooking also generates some economic spillover effects through the supply value chain when imported LPG is substituted with domestically produced electricity. Estimations of wider economic impacts are calculated based on data and information provided in **Table 15**. Note, however, that the estimations of wider economic benefits are rather crude because the fixed coefficients to express the relationships between variables, such as number of workers per MW of hydropower construction and operation may vary projects to projects depending on the project type. For example, run-of-river type hydropower projects require lower number of workers than reservoir type hydropower projects. The number of workers (jobs) required differ across projet sites. The wider economic benefits estimated here are more indicative rather than factual.

Table 15: Input data and assumptions for estimating economic spillover effects of substituting LPG with electricity for household cooking in Nepal (data are for the year 2023 unless specified otherwise)

Data item	Value	Source
Average capital cost of new hydro plant (US$/kW)	2500	Compiled from news articles providing hydropower costs by projects
LPG used for residential cooking (GWh)	5,048	WECS (2024a)
Electricity transmission & distribution loss (%)	12.73	NEA (2024)
Average construction period (year)	5.00	Project completion information from several hydropower projects
Hydropower plant capacity factor	57.65	Calculated using data from the Social Accounting Matix (SAM) 2018 prepared by Pradesha and Thurlow (2021) and revised by Timilsina et al. (2025). These values are based on 2018 SAM, as no more recent SAM is available.
Value added to the gross output ratio of the electricity sector	0.24	
Value added to the gross output ratio of the construction sector	0.27	
The ratio of labor input to gross output in the electricity sector	0.11	
Output multiplier of the construction sector	4.55	
The ratio of labor input to gross output in the construction sector	0.21	
Output multiplier of the electricity sector	3.53	
Average wage rate in the electricity sector ('000 NRs)	1433.8	Calculated using data from the 2022 Labor Market Report (DTUDA, 2024) and Nepal Labor Survey Report, 2017/18 (CBS, 2019)
Average wage rate in the construction sector ('000 NRs)	968.4	
NEA's labor payments per unit of electricity sale (Million NRs per GWh)	0.59	Calculated using ESMAP data (World Bank, 2025a)
Exchange rate (NRs/US$)	120.00	World Development Indicator Database (World Bank, 2025b)
GDP (Million US$)	39,618	39,618 World Development Indicator Database (World Bank, 2025)

Substituting LPG with electricity for cooking in Nepal, particularly through the expansion of hydropower infrastructure, offers significant macroeconomic benefits. This transition can lead to a substantial reduction of the import bill on LPG. Moreover, LPG is cross-subsidized by other petroleum products imported. The substitution of LPG with electricity would also reduce the cross-subsidy. In 2023, Nepal consumed 18 PJ of LPG for household cooking (see Table 2), equivalent to 5,048 GWh (see Table 7). Assuming an average capacity utilization factor of 57.8% for hydropower plants in Nepal and a transmission and loss rate of 12.73% in

2023, the country would have needed 1,145.4 MW of additional hydropower capacity. It would require a total of US$2,863.5 million investment for the construction of hydropower plants. Assuming a five-year construction period, the annual investment needed (in constant terms) would be US$ 590 million. The construction of hydropower would create approximately 14,600 jobs per year during the five-year construction period. It will create an additional 2,000 jobs each year throughout the power plant's operation. These numbers may seem surprising, but the total electricity required to replace the amount of LPG consumed in 2023 is approximately half of the total electricity consumed in the country that year. To generate one unit of output from the construction sector, it pays 0.21 for labor (Pradesha and Thurlow, 2021). The average annual wage payment to construction workers is US$8,000, which is significantly lower than in many countries around the world (DTUDA, 2024; CBS, 2019).

The construction of hydropower power plants with a total capacity of 1,145.4 MW would create US$157 million of value added per year during the construction phase. This represents approximately 0.4% of Nepal's GDP in 2023. Similarly, the operation of the power plants built to replace the LPG is estimated to generate US$ 124.4 million per year (or 0.31% of 2023 GDP) for the economic life of the power plants. These are the direct benefits of replacing imported LPG with domestically produced hydropower. Based on the Social Accounting Matrix (SAM) of Nepal for 2018 prepared by Pradesha and Thurlow (2021) and revised by Timilsina et al. (2025), generating one unit of output in the construction sector would generate 4.5 units of output throughout the economy. Similarly, generating one unit of output in the electricity sector would generate 3.5 output throughout the economy. Detailed data used for assessing economic spillover effects are provided in **Table 15**.

It could be argued that Nepal needs to import electricity during the dry season. An assumption of replacing LPG with domestic electricity might be unrealistic. However, the generation capacity is rapidly increasing, currently it stands more than 3,500MW[24]. Nepal is commissioning several power plants in near future to make the country self-sufficient in meeting its demand during the dry season. During the wet season it already has surplus capacity. The assumption here is that by the time Nepal is fully replace LPG for cooking, there would be enough capacity installed to meet the peak load (including the new cooking load) during the dry season.

[24] Department of Electricity Development. https://doed.gov.np/license/57

4.5 Sensitivity analysis

We also conducted a sensitivity analysis on the private costs of cooking, using fuel prices as the major component of the levelized cost. The economic analysis indicates that fuelwood cooking in rural areas is the most economical option in terms of total private costs, followed by electricity and LPG cooking. However, if the fuelwood price increases by 25%, cooking with electricity and LPG becomes cheaper in Kathmandu Valley (**Figure 5**). It is also interesting to note that even if the current fuelwood price were to double (i.e., a 100% increase), fuelwood would remain the cheapest option in all rural areas while cooking with electricity becomes a viable option in many urban areas (**Figure 5**). With fuelwood prices having tripled (a 200% increase), cooking with LPG has become a viable option in all urban areas and Nepal.

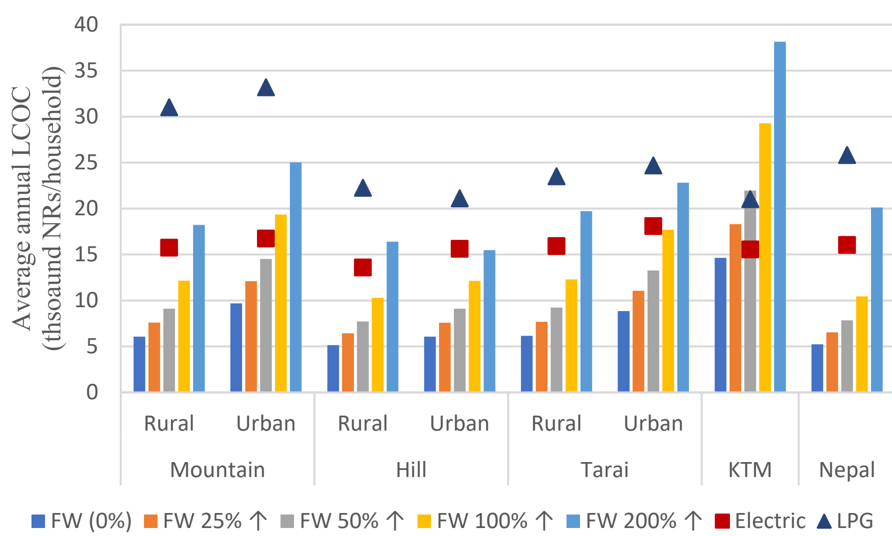


Fig. 5. Total annual levelized cost of cooking with varying fuelwood prices compared with electricity and LPG cooking by region

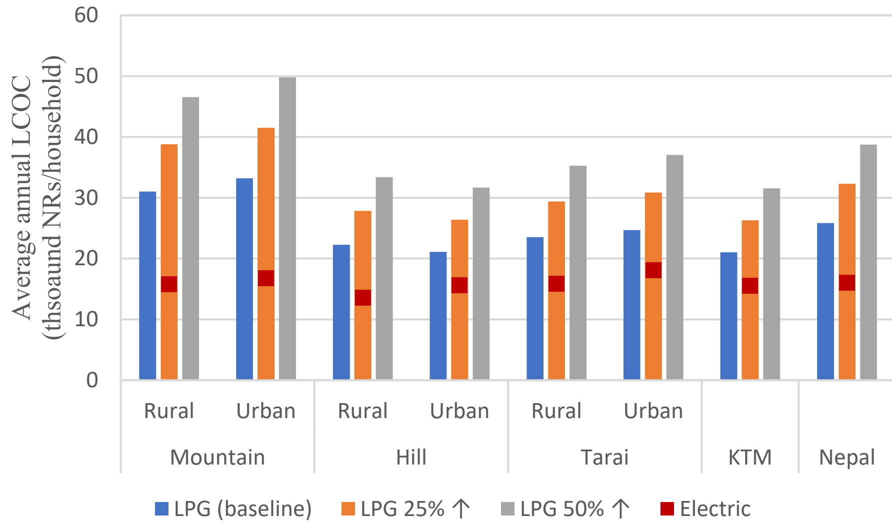


Fig. 6. Total annual levelized cost of cooking with varying LPG prices compared with electricity by region

We also find that cooking with LPG is more expensive than cooking with electricity. Currently, the total cost of cooking with LPG is higher than that of cooking with electricity nationwide. When the LPG price is increased by 50%, the total cost of cooking with LPG is about three times the cost of cooking with electricity in the mountain region and more than twice the cost of cooking with electricity in the hill and Tarai regions (**Figure 6**). Despite the lower cost of cooking with electricity, fuelwood in rural areas and LPG in urban areas are commonly used for cooking mainly due to a lack of electricity infrastructure, the high cost of electric cooking appliances, and a lack of awareness. These are the key areas that government policies should focus on to make electricity available and affordable to households, particularly in rural areas across the country so that they can switch from fuelwood to electricity.

5. Conclusions and Policy Implications

This study presents an economic analysis of various alternatives for transitioning from fuelwood and LPG to electric cooking in the residential sector from both private and social perspectives. The economic analysis finds that total cooking costs vary across regions due to input variables (e.g., cost of fuels, cooking device costs, and maintenance). The total cost of cooking from individual fuel sources varies widely by region and fuel/technology types, ranging from zero cost for TCS in rural areas to NRs 21.4 thousand for LPG in rural mountain regions for each household annually. In the case of more efficient and clean electric cooking, the annual total cost of cooking for each household varies from NRs 9 thousand for electric pressure cookers to 11 thousand for induction stoves. When fuel stacking is considered, the cost of cooking is the highest for households in the urban mountain and the Kathmandu Valley, at approximately NRs 20.5 and 17.9 thousand per year, respectively. In contrast, it is the lowest for households in the rural hill region, at approximately NRs 10.3 thousand per year. This result is expected because most urban households use LPG, which is relatively expensive compared to fuelwood in terms of cooking fuel costs per household.

Household air pollutants, particularly $PM_{2.5}$ from the combustion of unprocessed biomass fuels for cooking, pose a significant health challenge in rural areas of Nepal as it is estimated to have caused 25,069 deaths in 2019 and incurred NRs 40 billion in costs, equivalent to about 1% of the country's GDP. With fuel stacking, the external cost analysis reveals that the social cost of PM2.5 varies significantly, ranging from NRs 58 thousand per year for households in rural mountain regions to NRs 3 thousand for households in the Kathmandu Valley. It is the opposite of the social costs of CO_2 emissions, with lower social costs for rural households than for urban households.

With individual cooking fuels, the study finds that even if the current fuelwood price is doubled (i.e., a 100% increase), fuelwood remains the cheapest option in terms of the private cost of cooking in all rural areas, while cooking with electricity becomes a viable option in urban areas. With fuelwood prices having tripled (a 200% increase), cooking with LPG has become a viable option in all urban areas. However, cooking with LPG is more expensive than cooking with electricity and fuelwood, and increasing its price further increases the cost of LPG cooking.

The use of electricity for cooking is not only beneficial from the user's perspective but also from a national perspective. Since LPG is entirely imported and cross-subsidized, substituting LPG with electricity for cooking reduces import bills, improves Nepal's trade deficit with India, and creates jobs, generating value added to the economy.

The study's results suggest several policy implications. Since electricity is not only cleaner but also is significantly cheaper than LPG for cooking, electric cooking should be promoted. Evidence-based awareness is critical, as an individual consumer may not have the capacity to conduct economic analysis and realize that electric cooking is a more cost-effective option for them compared to LPG. Informing consumers with the new knowledge gained from analyses like this should be prioritized. If the cost difference between electric and LPG cooking is not enough for some consumers for any reason, such as behavioral factors or pricing instruments (i.e., an environmental fiscal policy such as a carbon tax), it can be used to make electric cooking more economical compared to LPG for cooking. Manufacturing electric cookstoves within the country and enabling the markets for electric cookstove maintenance would make electric cooking readily available and affordable. Reducing the initial costs by providing financial arrangements that allow consumers to pay the capital costs in installments over time, rather than the total cost at the time of purchase, could also help facilitate the deployment of electric cooking. Offering incentives for the local manufacturing of electric stoves and appliances further enhances cost reduction and encourages the adoption of electric cooking. Public awareness and education campaigns highlighting the health benefits, as well as the cost savings, would further promote the adoption of electric cooking.

Access to electricity alone would not be sufficient for promoting electric cooking, electricity supply must be reliable. Currently, power supply reliability is a critical barrier to the rapid adoption of electric cooking in Nepal. A large investment is needed to strengthen and expand the electricity distribution systems (e.g., replacing/upgrading of distribution transformers, distribution wires and poles, wider provision of smart electric meters) so that reliable supply of electricity can be secured to encourage electic cooking in Nepal.

This study is limited to a simplified static analysis while estimating the economic spillover effects of electric cooking. A natural expansion of the study could be the assessment of the power sector and overall economic impacts of scaling up electric cooking under alternative scenarios for the future growth of cooking energy and gradual replacement of LPG and other fuels with electricity in the residential as well as the commercial sector (e.g., hotels, restaurants, schools).

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Appendix A: Additional Data and Assumptions

Administratively, Nepal is divided into seven provinces. Each province is further divided into districts, and these districts are subdivided into urban and rural municipalities. The recent census data shows that these seven provinces have distinctive household structures and a wide range of types of fuel usually used for cooking (**Table A.1**). For instance, the centrally located three provinces (Bagmati, Gandaki, and Lumbini) together represent 51% of all households, followed by the eastern two provinces (Koshi and Madhesh) at 35%, and by western two provinces (Karnali and Sudurpachim) at 14%. More than half of all households in each province are in urban areas, with the highest rate (77%) in Bagmati, which includes the capital city Kathmandu Valley.

Table A.1: Province-level household structure and their main source of cooking fuel in 2021

Province	Household (thousand)	Urban (%)	Share of households by type of fuel usually used for cooking (%)								
			Urban			Rural			National		
			FW	LPG	ELE	FW	LPG	ELE	FW	LPG	ELE
Koshi	1,192	62	42	54	0.5	73	18	0.5	53	41	0.5
Madhesh	1,157	73	57	33	0.8	64	17	0.9	59	29	0.8
Bagmati	1,571	77	15	84	0.7	78	21	0.1	29	70	0.6
Gandaki	662	66	33	65	0.2	74	24	0.1	47	52	0.2
Lumbini	1,142	55	41	55	0.4	70	26	0.4	54	42	0.4
Karnali	366	52	72	27	0.1	94	5	0.1	82	17	0.1
Sudurpachim	577	62	60	35	0.3	87	9	0.2	70	26	0.3
Nepal	6,667	66	39	57	0.6	75	19	0.4	51	44	0.5

Note: FW refers to fuelwood, and ELE refers to electricity. Source: NSO (2023).

At the national level, the common features of cooking patterns include a high proportion of fuelwood (over 47%) as the main source of cooking fuels in six provinces except for Bagmati. In comparison, it is LPG (over 41%) in four provinces (Koshi, Bagmati, Gandaki, and Lumbini). IN Bagmati, about 70% of households rely on LPG as their main source of cooking fuels, the highest among all provinces, followed by fuelwood (29%) and electricity (0.6%). Households in the relatively less developed two western provinces rely heavily on fuelwood as their main source of cooking fuels, from 70% in Sudupachim to 82% in Karnali. Overall, more than half of all households rely on fuelwood as their main source of cooking fuels, followed by LPG (44%) and electricity (0.5%). The proportion of households using animal dung and biogas as their main source of cooking fuel is very small and varies across the provinces. However, in most provinces, the number of households relying on animal dung and biogas as their primary source of cooking fuel is higher than those relying on electricity. For example, households' reliance on animal dung ranges from less than 0.1% in Bagmati to as high as 11% in Madhesh. In comparison, biogas ranges from a low of 0.6% in Bagmati to as high as 3.4% in Sudurpachim (NSO, 2023).

In urban areas, LPG is the most common source of main cooking fuels. It ranges from a high of 84% in Bagmati to a low of 27% in Karnali. Interestingly, more than one-third of all urban households still rely on fuelwood as their primary source of cooking fuel. In some urban areas in less developed western provinces, households' reliance on fuelwood is as high as 72% (Karnali). Electricity for cooking in urban areas remains low, less than 1%, in all provinces. As expected, households in rural areas heavily rely on fuelwood as their main source of cooking fuel, ranging from 64% in Madhesh to 94% in Karnali. In comparison, their reliance on LPG ranges from 5% in Karnali to 26% in Lumbini.

Table A.2: Household size and useful energy requirement for cooking by region in Nepal

	Mountain		Hill		Tarai		Kathmandu valley	Nepal
	Rural	Urban	Rural	Urban	Rural	Urban		
Household size (person/hh)	4.38	4.23	4.14	3.90	4.99	4.63	3.89	4.37
Useful energy* (GJ/per/year)	0.890	0.988	0.791	0.988	0.791	0.988	0.988	0.912
Cooking events (Index, Kathmandu = 100)**	90	100	80	100	80	100	100	92

Note: * The annual per capita useful energy (0.988 GJ) is estimated based on the weighted average useful energy per capita of four cities (Biratnagar, Kathmandu Valley, Pokhara and Butwal) taken from WECS (2024a). ** The no. of household heating/cooking events for Kathmandu Valley is 7.9 per day (Index, 100) taken from Poudel et al. (2023). We assume 20% fewer cooking and heating events in rural areas of the Tarai and Hill regions and 10% fewer in rural areas of the mountain regions compared to the Kathmandu Valley. All urban areas experience the same cooking and heating events as those in the Kathmandu Valley. The estimated useful energy per capita for each region is calculated by multiplying 0.988 GJ/capita/year by the corresponding cooking and heating event index. The household size data is taken from (NSO (2023).

Based on the useful energy (Table A.2) and device efficiency (Table 8), the estimated annual final energy consumption of households using different cookstoves for cooking across the region is calculated and summarized in Table A.3.

Table A.3: Household's annual final energy consumption by region across the country (GJ/household/year)

Cookstove/fuel combination	Mountain		Hill		Tarai		Kathmandu Valley	Nepal	
	Rural	Urban	Rural	Urban	Rural	Urban		Rural	Urban
Traditional cookstove	32.5	34.9	27.3	32.2	32.9	38.1	32.0	29.7	35.5
Improved cookstove	16.2	17.4	13.6	16.1	16.4	19.1	16.0	14.9	17.7

Mettalic cookstove	21.6	23.2	18.2	21.4	21.9	25.4	21.3	19.8	23.7
Biogas	8.1	8.7	6.8	8.0	8.2	9.5	8.0	7.4	8.9
Kerosene	8.7	9.3	7.3	8.6	8.8	10.2	8.5	7.9	9.5
LPG	7.1	7.6	5.9	7.0	7.2	8.3	7.0	6.5	7.7
Hot plate	5.6	6.0	4.7	5.5	5.6	6.5	5.5	5.1	6.1
Infrared	5.6	6.0	4.7	5.5	5.6	6.5	5.5	5.1	6.1
Induction	5.0	5.4	4.2	4.9	5.1	5.9	4.9	4.6	5.5
Electric pressure cooker	4.5	4.9	3.8	4.5	4.6	5.3	4.5	4.1	5.0

Note: We assume fuelwood for traditional, improved, and metallic cookstoves.